

- According to Miller Manufacturing, a hen needs at least 1 pint of water a day.
 - This information was used in deciding how big our barrel needs to be.
 - The barrel we are using is 55 gallons (440 pints) divided out to 11 hens
- According to Carolina Farm Stewards, chickens can drink rain water.
 - Knowing this information we moved forward with our idea of the rain collection barrel.
- We used math to measure lengths of tubing/piping and the placement of the barrel.
- Weatherspark.com - Owings Mills MD weather. It is important to document how much water the area gets in order to determine there will be enough water for the chickens in a set time period. If there is not enough rainfall for all chickens, humans may still need to interact or we need to find another solution. Other factors play into the water collection like humidity, evaporation, and precipitation. We need to be thoughtful of weather conditions and the many factors that play into rainfall and water collection. A basic understanding of weather is needed.
 - May has the most rainfall with an average of 3.5 inches
 - January has the least rainfall with an average of 1.9 inches
 - The wet season lasts 4.8 months March 31 to August 24
 - The dry season lasts 7.2 months August 24 to March 31
 - The most days of rain is in June
 - Most precipitation in July
 - One of the goals of our product is to have enough water for the 7 chickens (now 5 chicks) during the summer. This is in case no one is on campus during the summer to water the chickens.
- According to bestplaces.net, Owings Mills has an annual average of 43 inches over a 114 day period.
- According to weatherandclimate.com Owings Mills 47.54 rainy days
- Our experts are Ms. Shao and Ms. Harrison. They both have experience with tending to chickens. Ms. Shao teaches biology and tended to chicks in her classroom. Ms Harrison herself has chickens at home she takes care of so she is very knowledgeable.
 - Once we brought our idea and planned prototype to Ms. Shao, she approved and added the suggestion to have a pipe inside the coop the chickens can directly drink from. We added that idea to our new prototype. Before this addition to our prototype, we would've had to pour water into the existing gallon in the middle of the coop (less effective).
- According to The Chicken Chick, water nipples are really resistant to bacterial growth unlike open waterers that can easily become dirty.